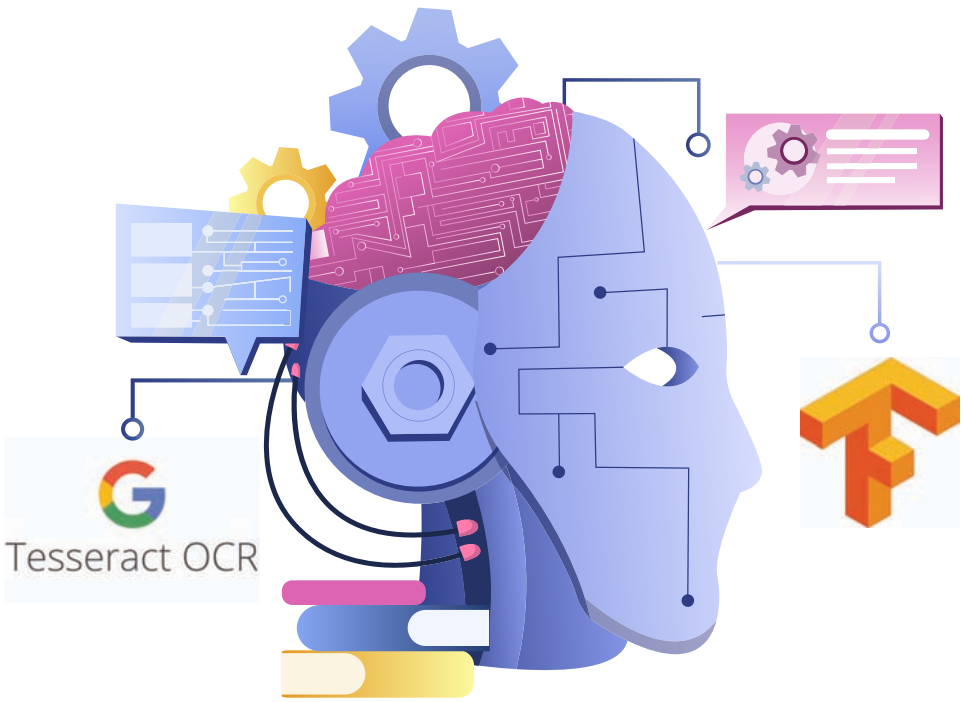
TensorFlow and Tesseract OCR: Two Popular AI/ML Tools



The article gives an overview of two popular AI and ML tools — TensorFlow and Tesseract OCR — which have been developed by Google AI Labs for the open source community.

Google AI Labs provides many services for AI and ML. These include the use of free platforms for development activities, releasing code to the open source community, and support for AI/ML related research activities.

TensorFlow is a Google AI project and one of the most popular open source machine learning frameworks. It can be used to build and train ML models like Keras API. It builds neural networks, and enables machine translation and video processing using ML models.

Tesseract OCR is another popular open source character recognition and OCR library written in Python and C++. It was originally developed as a commercial OCR package by HP Laboratories in late 1990 to run on DOS command-line mode and later to work on the Windows OS by enhancing features using C++. In 2005, it was released by HP Laboratories as an open source framework available for community development in GitHub, and was sponsored by the Google AI project for this. It is still available under Apache License 2.0 for open source applications.

TensorFlow and its features

TensorFlow is built on neural network models and supports many popular algorithms like clustering, decision tree, neural network evaluation, linear regression, and Naïve Bayes classifier. It is very popular for large volume machine learning activities and supports both CPU and GPU based processing, which can help to optimise operational costs in building machine learning solutions.

There are many frameworks that support building a machine learning model, and their flexibility of usage varies. For example, Scikit-Learn has many inbuilt libraries for machine learning algorithms and implementation, whereas TensorFlow is more flexible as it allows development and implementation of custom ML